



Submission Title: Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task

SUBMISSION PREVIEW: VISUAL-HAPTIC FEEDBACK ENHANCES FINGER INDIVIDUATION IN A VIRTUAL PRECISION GRIP NEUROTRAINING TASK

[Visual-haptic feedback enhances finger individuation in a virtual precision grip neurotraining task](#)

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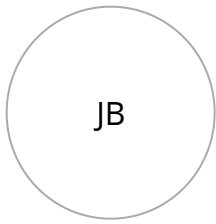
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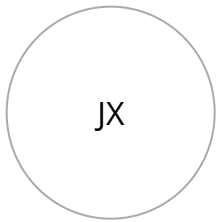
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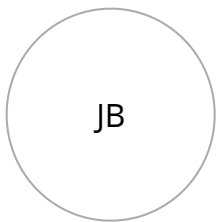
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Abstract

Track/Category

Orthopedic and Rehabilitation Engineering

Sub-track:

Prosthetics, Devices, and Therapies for Rehabilitation of Orthopaedic Tissues

Introduction:

Independent control of each finger and coordination among them are fundamental aspects of human hand dexterity. A common consequence of stroke is the loss of finger dexterity. Restoring precise force application in a precision grip is vital for regaining effective hand function in daily activities post-stroke, making it a key focus in rehabilitation research. The ultimate goal of our study is to examine the role of visual and haptic feedback in the reacquisition of this lost finger individuation and coordination abilities in rehabilitation. Utilizing our previously developed finger force measurement device augmented with a cutaneous tactile device at the fingertips, we asked healthy participants to complete a virtual precision grip task in four conditions that modulated the availability of haptic and visual feedback. The following pilot results serve to validate our device and provide baseline performance measurements for future studies with stroke patients.

Materials and Methods:

The study utilized the Hand Articulation Neuro-training Device (HAND) to capture isometric forces from the thumb and index fingertips of the right hand (Fig. 1). The HAND includes strain gauge assemblies at each fingertip, measuring forces with 2.56mN resolution in 3-DOF at 120Hz. Applied forces were mapped to cursors in a virtual environment. Voice coil actuators attached to the device provided cutaneous haptic feedback, with vibration amplitude linearly increasing with force applied to the virtual object. Additionally, visual feedback was provided, wherein the brightness of the virtual object linearly increased with the force applied to the virtual object.

Eight participants with no upper-limb impairments or history of neurological or movement disorders were recruited to use the HAND to precisely pinch and hold (within 5N force) a virtual sphere for 5 seconds within a 20 seconds time window, guided by visual and haptic feedback (Fig. 2). Two different sphere sizes and control-display ratios were utilized in a randomized and counterbalanced manner to encourage a reliance on the provided feedback. The experiment was structured into six phases: an initial training set of 12 trials with combined visual-haptic feedback, followed by 12 trials without feedback to establish a baseline. Subsequently, participants completed 24 trials, each of visual, haptic, and visual-haptic feedback conditions, in blocks of 8 trials with the order counterbalanced across participants. The session concluded with 12 trials without feedback. High-resolution 3D isometric forces were recorded during each trial.

Results, Conclusions, and Discussions:

The study's results indicate a clear hierarchy in the efficacy of feedback modalities in enhancing task performance in a virtual precision grip task (Fig. 3). A Kruskal-Wallis test revealed significant differences in the percentage of successful trials across the feedback conditions ($\chi^2(4)=17.12$, $p=0.0013$). The combined visual-haptic feedback condition yielded the highest number of successful trials (60.3% success), outperforming the haptic-only (56.1% success) and visual-only (34.5% success) feedback conditions. This result suggests that the integration of multiple sensory pathways is more beneficial for motor learning and precision in tasks requiring fine motor control.

Haptic feedback may be more effective than visual feedback in a precision pinch-and-hold task because it directly corresponds to the sense of touch, which is primarily used to gauge interaction forces. Estimating force visually, even with visual feedback in the virtual space, is challenging and often less precise. The haptic sensory loop is also faster than the visual loop. The visual feedback condition alone showed better performance than the no-feedback condition, highlighting the importance of sensory cues in motor tasks.

Further analysis focusing on individual participant's performance before and after the feedback tasks confirmed that the training effect though seen, was not significant ($\chi^2(1)=3.40$, $p=0.0652$). Participants

performed better in the post-training no-feedback condition (18.6% success) than in the pre-training phase (5.3% success), indicating positive transfer of skills acquired during feedback-mediated training to contexts where no sensory feedback was available.

These findings support our hypothesis that visual-haptic feedback enhances immediate motor execution and contributes to better retention and transfer of the learned finger dexterity skills, potentially due to more effective engagement of sensorimotor loops involved in motor memory and learning. This has profound implications for designing rehabilitation protocols and devices. Our study emphasizes the utility of visual and haptic feedback in neurorehabilitation and motor learning contexts, suggesting that multimodal sensory feedback could potentially enhance motor recovery in individuals with sensorimotor impairment due to neurological injuries. Future studies will explore the neural correlates of these findings and consider longitudinal assessment to probe the long-term benefits of visual and haptic augmented feedback in stroke survivors.

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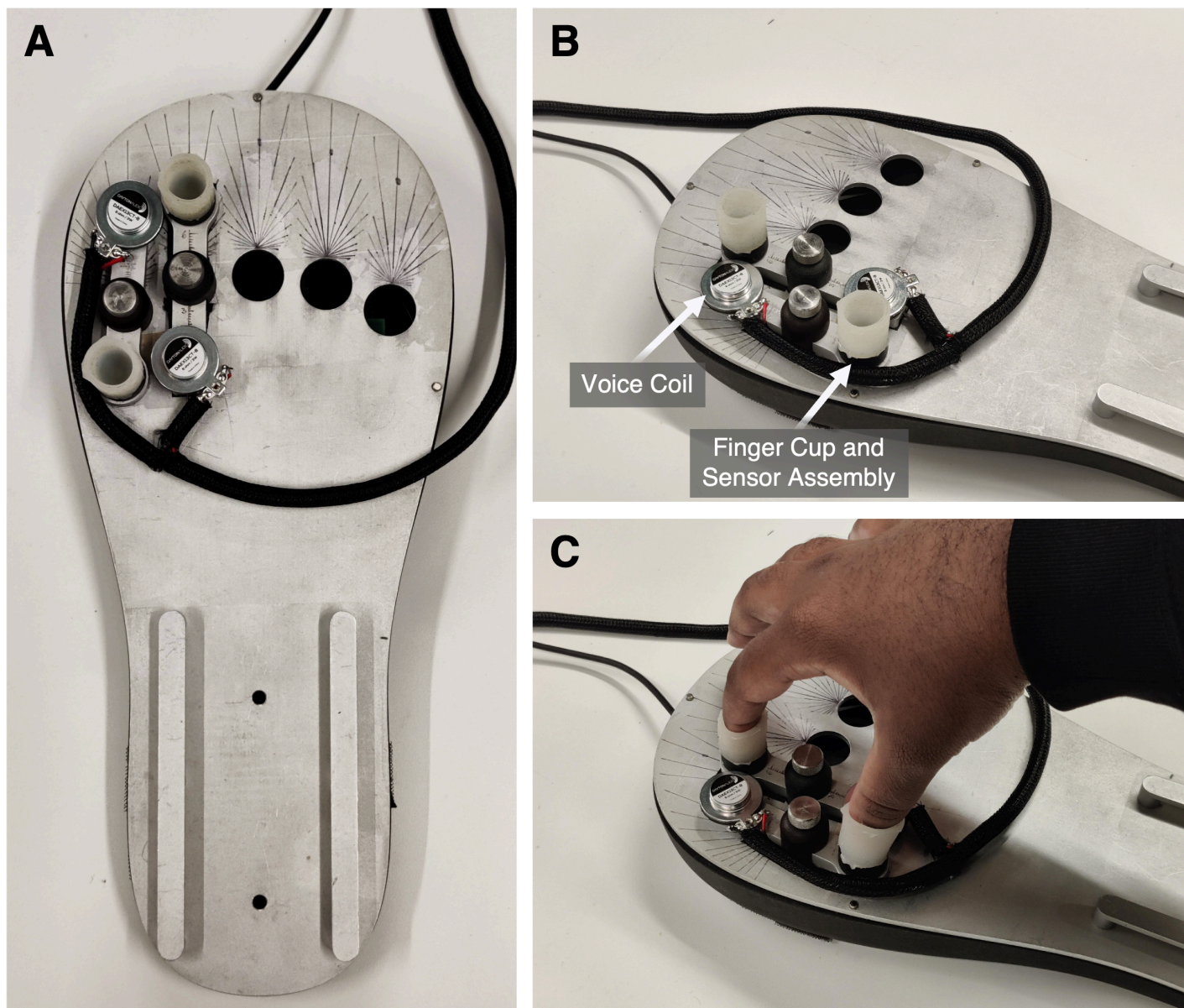


Figure 1. A. Hand Articulation Neuro-training Device (HAND) setup; B. Voice coil and sensor placement on the HAND; C. Participant using the HAND.

Figure_01.png

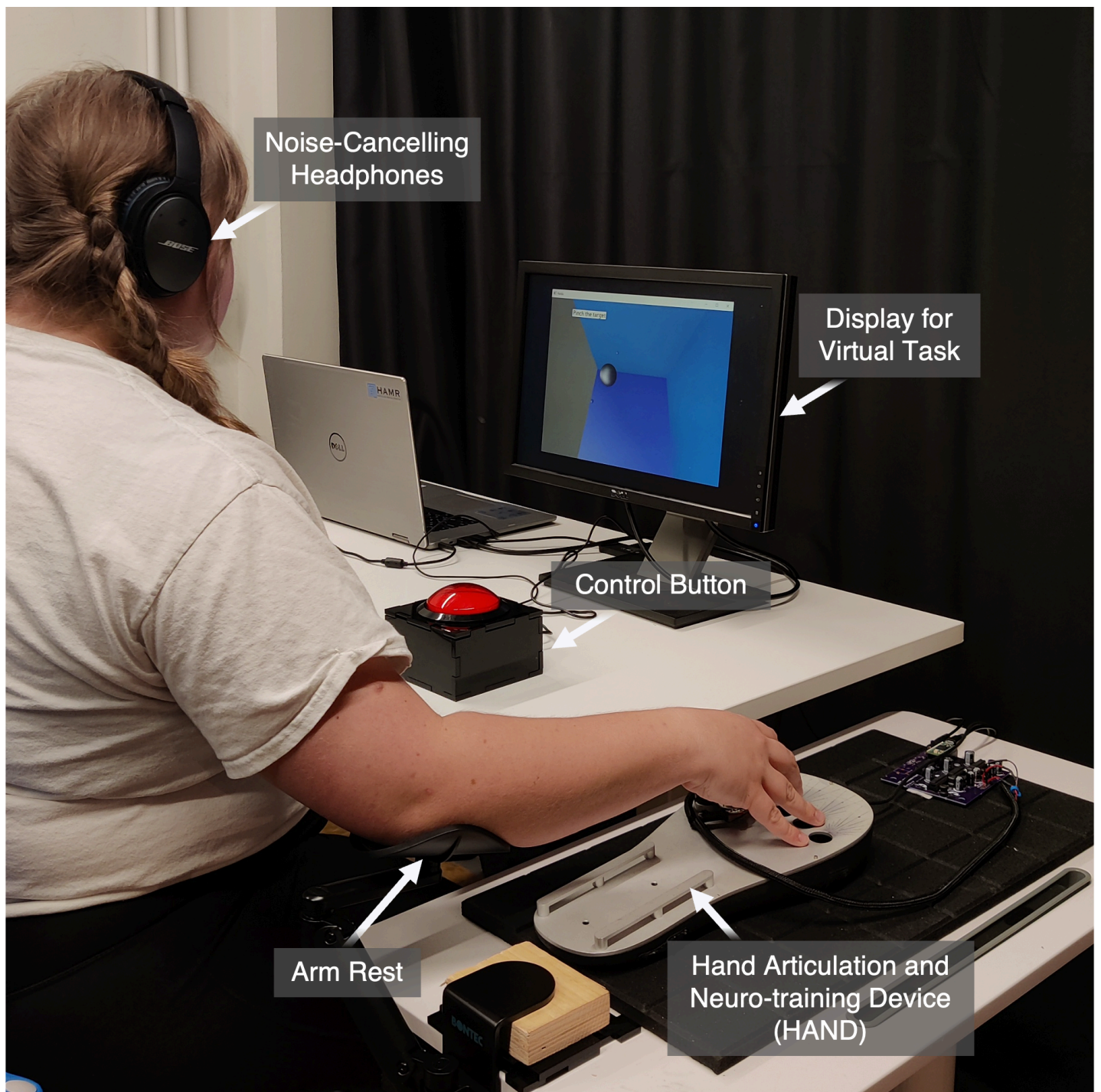


Figure 2. Experimental setup for data collection.

Figure_02.png

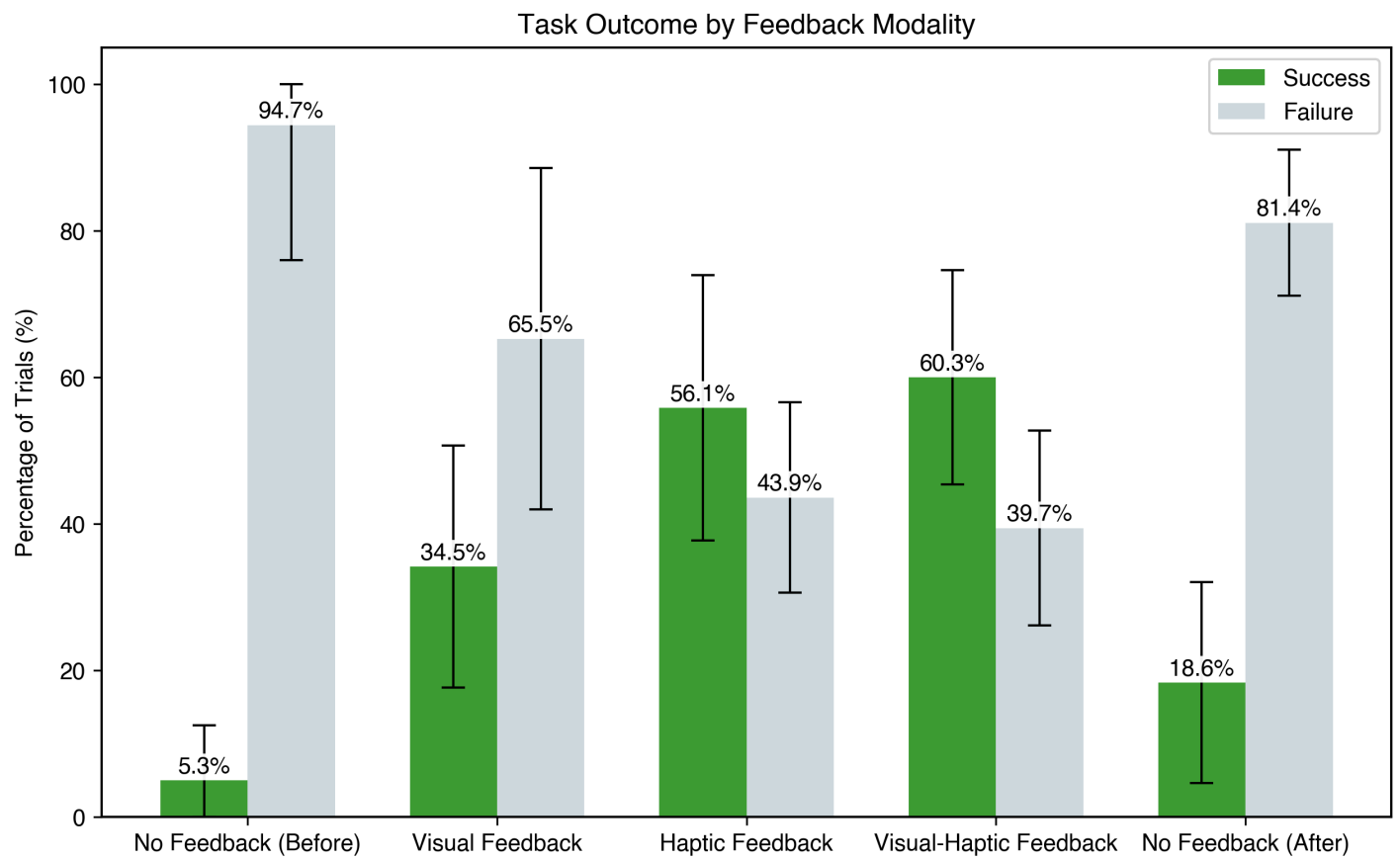


Figure 3. Summary of task outcomes grouped by feedback modality across all participants.

Figure_03.png